PHISHING DETECTION & AWARENESS REPORT

Email Threat Analysis & Security Awareness Guidelines

Based on analysis of 326 real-world email samples

Dataset Sources: Enron Corpus | Stanford | Lehigh | University of Washington

Prepared: 2025 | Classification: Internal Use

1. Executive Summary

This report presents a comprehensive analysis of phishing email threats based on a curated dataset of 326 email samples from verified public sources. The dataset — drawn from the Enron Corpus, Stanford University, Lehigh University, and the University of Washington — was used to identify, classify, and explain phishing techniques in business-friendly language.

Key findings from this analysis:

- 50% of all analyzed emails (163 out of 326) were confirmed phishing attempts
- Phishing emails commonly impersonated universities, IT helpdesks, and well-known brands (Apple, HR departments)
- COVID-19 themed lures and fake financial aid offers were among the most prevalent social engineering tactics
- Urgency language, suspicious links, and credential harvesting were the top three phishing techniques identified
- Legitimate (safe) emails showed natural language, organizational context, and no requests for credentials

This report aims to help employees recognize phishing threats, understand attacker techniques, and follow clear prevention guidelines to protect organizational security.

2. Dataset Overview

The analysis is based on a balanced, real-world email dataset with the following characteristics:

326	163 (50%)	163 (50%)	4
Total Emails Analyzed	Phishing Emails	Legitimate Emails	Data Sources

2.1 Data Sources

Indicator	Description	Risk Level
Enron Corpus	Legitimate corporate emails from the Enron legal proceedings; used as the safe email baseline	SAFE
Stanford Phishing	Phishing simulation examples from Stanford University IT Security	HIGH
Lehigh University	Documented phishing examples from Lehigh University security awareness program	HIGH
Univ. of Washington	Real phishing emails reported by UW community members	HIGH

3. Phishing Indicators Identified

Through analysis of all 163 phishing emails in the dataset, the following key indicators were identified. Understanding these patterns enables employees to spot threats before they cause harm.

Indicator	Description	Risk Level
Urgency & Pressure	Language like 'immediately', 'suspended', 'act now' designed to prevent careful thinking	**HIGH**
Suspicious URLs	Misspelled domains, double paths, or links that don't match the claimed sender (e.g., lehighedu.webs.com)	**HIGH**
Credential Requests	Asking users to 'verify', 'log in', or 'confirm' passwords or personal information	**HIGH**
Impersonation	Emails pretending to be from IT, HR, Apple, universities, or government agencies	**HIGH**
Grammar & Spacing Errors	Deliberate misspellings like 'me ss ages', 'dela.ye.d' to evade spam filters	**MEDIUM**
Generic Greetings	Using email addresses as names ('Dear user@stanford.edu') instead of real names	**MEDIUM**
Financial Lures	Fake offers of COVID relief funds, financial aid, or job opportunities	**HIGH**
Vague Sender Identity	'Admin Help Desk', 'Assistant Professor' without verifiable credentials	**MEDIUM**
Mismatched Branding	Copyright notices and logos that don't match the actual sender organization	**MEDIUM**
Unexpected Attachments	References to 'private documents' sent without prior context or relationship	**HIGH**

4. Phishing Techniques Explained

The following section explains the most common phishing techniques found in this dataset, written in plain language for a business audience.

4.1 Credential Harvesting

The most common technique. The attacker sends a fake login page link, typically mimicking a trusted organization like a university portal or Apple ID. When victims enter their username and password, the credentials are sent directly to the attacker.

Example from dataset (PHISHING):

Email Classification: PHISHING

```
Your e-mail access has been suspended for your security.
To regain your access Click here
Stanford University.
```

What makes this dangerous: Short, emotionally charged message. No verifiable sender. Generic 'Click here' link with no visible URL. No personal greeting.

4.2 Social Engineering with Urgency

Attackers create artificial time pressure to prevent victims from thinking critically. Common triggers include: account suspension warnings, storage quota exceeded, limited-time financial offers, and pandemic-related emergency notices.

Example from dataset (PHISHING):

Email Classification: PHISHING

```
WEBMAIL UPGRADE
Attention!!! Lehigh Email User,
This message requires that you verify your Webmail immediately. Our email lab
discovered You are currently using 2.5GB space instead of 2GB. You will be unable
to receive new email, loss important information in your mailbox or cause limited
access to it if you don't activate here.
To complete activation process, visit upgrade webpage below:
http://l...
```

Red flags: Triple exclamation marks. Fake quota warning. Suspicious external domain (webs.com is a free website builder, not Lehigh University). Grammar errors.

4.3 Brand Impersonation

Attackers impersonate trusted brands — universities, Apple, HR departments — to add credibility. They often copy logos, copyright notices, and formatting while using fraudulent links or requests.

Example from dataset (PHISHING):

Email Classification: PHISHING

```
Dear customer,
Your Apple ID was used to sign in to iCloud on an iPhone 4.
Time: February 06, 2014
Operating System: iOS;6.0.1
If you have not recently signed in to an iPhone with your Apple ID and believe
someone may have accessed your account, please click here to confirm your details
and change your password.
```

```
Apple Support
Copyright © 2014 iTunes S.à r.l.
```

Red flags: 'Dear customer' instead of your real name. Asks to 'click here' to confirm details. Real Apple would never ask for password confirmation via email link. Outdated iOS version reference.

4.4 Filter Evasion via Deliberate Misspellings

Some sophisticated phishing emails deliberately insert spaces, dots, or extra characters into words to bypass automated spam and phishing filters, while still being readable to human targets.

Example from dataset (PHISHING):

Email Classification: PHISHING — Filter Evasion

```
Hi user@stanford.edu,
Due to a se rv.er error on your e-mail
, (user@stanford.edu) (5) in coming me ss ages were d.ela.yed..
Log on to your port al to reco ver your dela.ye d messages
```

What to notice: Words like 'se rv.er', 'me ss ages', 'dela.ye d' are normal words deliberately broken up. Legitimate systems never send emails that look like this.

4.5 COVID-19 & Crisis Lures

During and after the COVID-19 pandemic, attackers heavily exploited pandemic-related themes. These emails offered fake financial aid, emergency funds, job opportunities, or health information to target anxious recipients.

Example from dataset (PHISHING):

Email Classification: PHISHING

```
The University of Washington System is sharing funds for all students during this
pandemic, please update your financial aid status to claim yours.
Login.uw.edu/covid-19-aid-update
For instructions on Accepting Your Financial Aid on
https://login.uw.edu/login/login./.
Regards,
Assistant Professor
University of Washington
```

Red flags: Suspicious nested login URL (login./). Vague sender title 'Assistant Professor'. Exploits pandemic anxiety to lower victim guard. No specific student name or ID.

5. Characteristics of Legitimate (Safe) Emails

Understanding what safe emails look like is just as important as recognizing phishing. The 163 legitimate emails in the dataset (from the Enron corporate corpus) shared these common characteristics:

Email Classification: LEGITIMATE (SAFE)

```
Subject: james valverde - interview schedule
attached you will find the interview packet for the above-referenced person. the
interview will happen thursday, february 1, 2001. please print all three documents
for your hard copies. if you have any questions, or conflicts of schedule, please
do not hesitate to contact me.
sasha divelbiss
58714
```

Why this email is legitimate:

- Clear, specific subject line referencing a real person
- Natural, conversational language without artificial urgency
- No requests for passwords, credentials, or personal information
- Sender provides their name and extension number — verifiable identity
- No suspicious links or unexpected attachments
- Organizational context is clear and consistent

5.1 Safe vs. Phishing: Quick Comparison

Feature	Legitimate Email	Phishing Email
Greeting	Uses your real name	Generic: 'Dear user' or email address
Sender	Known contact, verifiable domain	Vague title, suspicious domain
Tone	Normal, specific, conversational	Urgent, alarming, pressuring
Links	No links, or verifiable org links	Suspicious domains, 'Click here'
Request	No credential or payment requests	Asks for login, verification, payment
Language	Clear grammar and spelling	Errors, deliberate misspellings
Context	Refers to prior relationships	Unsolicited, out-of-nowhere contact

6. Prevention & Awareness Guidelines

The following guidelines are recommended for all employees and users to reduce the risk of falling victim to phishing attacks. These are organized by priority level.

Action	Description	Priority
Verify Before Clicking	Never click links in unexpected emails. Manually type known URLs into your browser instead of using email links.	**Critical**
Check the Sender	Examine the full email address, not just the display name. 'Apple Support' can be sent from any address.	**Critical**
Never Share Credentials	No legitimate organization will ask for your password via email — ever. Treat any such request as phishing.	**Critical**
Enable MFA	Multi-factor authentication prevents attackers from accessing accounts even if credentials are stolen.	**Critical**
Hover Over Links	Before clicking, hover your mouse over links to see the actual destination URL in the status bar.	**High**
Report Suspicious Emails	Use your organization's phishing report button or forward to your security team immediately.	**High**
Pause Before Reacting	Urgency is a manipulation tactic. If an email makes you panic, stop and verify through an independent channel.	**High**
Verify Phone Numbers Independently	If an email provides a callback number, look up the official number yourself instead of using the one provided.	**High**
Keep Software Updated	Outdated browsers and email clients are more vulnerable to malicious links and attachments.	**Medium**
Attend Security Training	Regular awareness training is the most effective defense against social engineering attacks.	**Medium**

6.1 The STOP-THINK-VERIFY Framework

When in doubt about any email, apply this three-step framework before taking any action:

STOP	THINK	VERIFY
Pause your reaction. Do not click, reply, or download anything until you have assessed the email carefully.	Ask: Am I expecting this? Does the sender make sense? Is this creating artificial urgency or pressure?	Confirm through an independent channel — call the sender directly or check the official website yourself.

7. Conclusions

This analysis of 326 real-world email samples confirms that phishing attacks are sophisticated, varied, and effective when users are unprepared. The key takeaways from this report are:

- Phishing makes up a significant portion of real-world email traffic — exactly 50% in this balanced dataset reflects the scale of the threat in practice
- The most dangerous phishing emails combine emotional manipulation (urgency, fear, greed) with technical deception (spoofed domains, fake branding)
- Legitimate emails are identifiable by their specificity, context, and absence of pressure tactics or credential requests
- Employee awareness is the single most important defense layer — technical filters alone are insufficient
- Simple habits — pausing before clicking, verifying senders, and reporting suspicious emails — dramatically reduce organizational risk

Organizations are encouraged to share this report with all staff, conduct regular phishing simulation exercises, and establish a clear reporting channel for suspected phishing attempts. Cybersecurity is a shared responsibility.

8. References & Data Sources

- Enron Email Dataset — Federal Energy Regulatory Commission (FERC) public disclosure
- Stanford University IT Security — https://uit.stanford.edu/phishing
- Lehigh University Security Awareness — https://lts.lehigh.edu/phishing/examples
- University of Washington CISO — https://ciso.uw.edu/education/more-phishing-examples/
- NIST Cybersecurity Framework — https://www.nist.gov/cyberframework
- Anti-Phishing Working Group (APWG) — https://apwg.org

This report was prepared for educational and security awareness purposes. All email samples are sourced from publicly available datasets used in academic and institutional security training programs.